Numerical methods in `dmplex.jl`

Meshes, compatible finite elements and C-grid finite volumes
for the rotating shallow-water equations on the sphere

Oswald Knoth

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Abstract

These notes collect the theory behind the discretizations implemented in `dmplex.jl`: the mesh generators and mesh optimizations (bisection, Voronoi duals, spring dynamics, centroidal Voronoi tessellations, cleanup of locally refined Delaunay meshes), the compatible finite-element models, the TRiSK finite-volume scheme on Voronoi meshes with the PV fluxes of Thuburn–Ringler and Gassmann, a kinetic energy that is both consistent and energy conserving, and the triangular C grid of ICON. For every scheme the conservation properties are derived and the known consistency defects are explained, together with the numerical evidence from Williamson's test case 2 and the Galewsky jet. The staggered kite elements have a separate note (`kite_shallow_water.tex`) and are only summarized here.

Contents

1 Equations and test cases **2**

2 Meshes **3**
2.1 Generators . . . 3
2.2 Cell centers and orthogonality . . . 3
2.3 Defects of bisection meshes . . . 4
2.4 Spring dynamics . . . 4
2.5 Spherical centroidal Voronoi tessellations . . . 4
2.6 Cleanup of locally refined Delaunay meshes . . . 4
2.7 Kite grids . . . 5

3 Compatible finite elements **5**
3.1 Spaces . . . 5
3.2 Linear shallow water . . . 5
3.3 Nonlinear shallow water . . . 6
3.4 Staggered kite elements . . . 6

4 Finite-volume upwind advection **6**

5 C-grid finite volumes on Voronoi meshes: TRiSK **6**
5.1 Geometry . . . 6
5.2 Operators . . . 7
5.3 The nonlinear TRiSK scheme . . . 7
5.4 Gassmann's PV flux . . . 8
5.5 Consistency of the reconstruction on optimized meshes . . . 8

6 A consistent, energy-conserving kinetic energy **8**
6.1 Why the cell kinetic energy is inconsistent 8
6.2 The vertex kinetic energy 9
6.3 Mixture and energy conservation 10
6.4 Results 10

7 The triangular C grid of ICON **11**
7.1 Operators 11
7.2 The kinetic-energy gradient on irregular triangles 12
7.3 Divergence damping 12
7.4 Locally refined MPAS meshes 12

8 Time integration **13**

9 Parallelization and GPUs **13**

10 Open questions **14**

1 Equations and test cases

On the unit sphere with outward normal $\hat{\mathbf{r}}$ the rotating shallow-water equations in vector-invariant form read

$$\partial_t \mathbf{u} + q\,\hat{\mathbf{r}} \times \mathbf{F} + \nabla(K + gh) = 0, \qquad \partial_t h + \nabla \cdot \mathbf{F} = 0, \tag{1}$$

with the mass flux $\mathbf{F} = h\mathbf{u}$, the kinetic energy $K = \frac{1}{2}|\mathbf{u}|^2$, the relative vorticity $\zeta = \hat{\mathbf{r}} \cdot \nabla \times \mathbf{u}$, the Coriolis parameter $f = 2\Omega z$ and the potential vorticity $q = (\zeta + f)/h$. Mass $\int h$, energy

$$E = \int hK + \tfrac{1}{2}gh^2 \tag{2}$$

and potential enstrophy $\frac{1}{2}\int hq^2$ are conserved. The derivation of energy conservation is the template for all discretizations below: with $\partial E/\partial \mathbf{u} = h\mathbf{u} = \mathbf{F}$ and $\partial E/\partial h = K + gh =: B$,

$$\dot{E} = \int \mathbf{F} \cdot \partial_t \mathbf{u} + B\,\partial_t h = -\int q\,\mathbf{F} \cdot (\hat{\mathbf{r}} \times \mathbf{F}) - \int \mathbf{F} \cdot \nabla B - \int B\,\nabla \cdot \mathbf{F} = 0,$$

since $\mathbf{F} \cdot (\hat{\mathbf{r}} \times \mathbf{F}) = 0$ pointwise and the gradient is the negative adjoint of the divergence. A discretization conserves energy if it keeps these three ingredients: an antisymmetric Coriolis (PV flux) operator, adjoint gradient and divergence, and a kinetic energy and a mass flux that are linked by $\partial E_h/\partial \mathbf{u} = \mathbf{F}$.

Units: radius $a = 1$, $\Omega = 1$ ($f = 2z$), and in the test cases $g = 1$ (TC2) or the nondimensional gravity of the Galewsky case.

Williamson test case 2 (TC2). Steady geostrophic solid-body rotation, $\mathbf{u} = u_0(-y, x, 0)$, $h = 1 - (u_0 + u_0^2/2)z^2$, $u_0 = 0.2$, $g = 1$. The height error after one time unit measures the consistency of the balance between the Coriolis term and $\nabla(K + gh)$. The maximum error is the sensitive quantity: an $O(1)$ inconsistency of an operator gives an $O(1)$ local truncation error of the momentum equation next to the mesh defects, which shows up as a maximum error that stops converging.

Galewsky jet. The barotropically unstable mid-latitude jet of Galewsky et al. (2004), $u_{\max} = 80\,\mathrm{m\,s^{-1}}$, mean depth 10 km, with the height perturbation at 45°N, integrated for 6 days without explicit dissipation (unless stated). It tests nonlinear stability: grid-scale noise, grid imprinting (wavenumber-5 pattern of the icosahedron) and the vortex roll-up from day 4.

2 Meshes

All meshes are `Plex` objects (DMPlex-like Hasse diagrams of cells, edges and vertices with cones and supports) on the unit sphere or on a doubly periodic plane. Cells are oriented counterclockwise seen from outside.

2.1 Generators

Icosahedral triangles (`create_spherical_mesh(n)`). The icosahedron is bisected n times: every triangle is split into four, the edge midpoints are projected radially onto the sphere. Level n has $20 \cdot 4^n$ triangles and $10 \cdot 4^n + 2$ vertices.

Hexagons (`create_hexagon_mesh(n)`). The spherical Voronoi diagram of the icosahedral vertices: the circumcenters of the triangles become the vertices, every triangle vertex becomes a cell (12 pentagons, the rest hexagons). The result is an exact Voronoi–Delaunay pair; the generators are the icosahedral vertices.

Cubed sphere (`create_cubed_sphere_mesh(n)`). Equiangular gnomonic projection (Sadourny): on every panel the node (i, j) is at the angles $\alpha_i, \beta_j = -\pi/4 + \pi i/(2n)$, mapped to $(1, \tan\alpha, \tan\beta)$ and normalized.

HEALPix (`create_healpix_mesh(nside)`). The corners of the $12\,\mathrm{nside}^2$ HEALPix pixels (continuous version of `xyf2loc`), connected by great-circle arcs; the areas are close to but not exactly equal.

Periodic planes (`create_periodic_plane`). Squares or triangles on a torus.

Grid files (`read_grid`). ICON triangle grids (`vertex_of_cell`) and MPAS Voronoi grids (`verticesOnCell`); with `dual=true` the hexagonal dual of an ICON grid (cells around its vertices, vertices at the triangle circumcenters) or the Delaunay triangles of an MPAS grid (vertices at the generators, triangles from `cellsOnVertex`).

2.2 Cell centers and orthogonality

Staggered schemes need a mass point per cell, and C-grid schemes need the dual edge (through the mass points of two neighbouring cells) to be orthogonal to the primal edge. `cell_centers` chooses

- the circumcenter for triangles (the vertex of the Voronoi dual);
- for Voronoi meshes the generators, recovered from the mesh alone: an edge with great-circle normal $\mathbf{n}$ between cells i, j lies on the perpendicular bisector of the generators iff $g_i - g_j \parallel \mathbf{n}$ and $\mathbf{n} \cdot (g_i + g_j) = 0$. These conditions are linear and homogeneous in all generators; their least-squares null vector is found by inverse iteration;
- the normalized vertex mean otherwise (cubed sphere, HEALPix).

The *orthogonality defect* is the largest $|\cos|$ of the angle between a primal edge and the great circle through the two centers; it is zero for a Voronoi–Delaunay pair, which the C-grid models require.

2.3 Defects of bisection meshes

Plain bisection keeps the shape of the 20 icosahedron faces: across the edges of the coarser levels the cell shapes jump, and these jumps do not shrink under refinement. A convenient measure for the triangular C grid is the asymmetry of the circumcenters about an edge, $|d_{e,l} - d_{e,r}|/d_e$, with $d_{e,l}$, $d_{e,r}$ the distances of the two circumcenters to the edge and $d_e = d_{e,l} + d_{e,r}$. It is 0.193 at every bisection level (ICON's R02B04/B05 grids, which are optimized: 0.15/0.12). On the Voronoi side the TRiSK tangential reconstruction has an $O(1)$ maximum error of 0.11 on plain hexagons at all levels (Section 5).

2.4 Spring dynamics

Following Tomita et al. (2002), the edges of the triangle mesh become springs of natural length $\beta\bar{d}$; the force on vertex i is

$$\mathbf{f}_i = P_i \sum_{j\sim i} (|\mathbf{x}_j - \mathbf{x}_i| - \beta\bar{d}) \, \frac{\mathbf{x}_j - \mathbf{x}_i}{|\mathbf{x}_j - \mathbf{x}_i|}, \qquad P_i = I - \mathbf{x}_i \mathbf{x}_i^T, \tag{3}$$

and the damped motion $\mathbf{v}_i \leftarrow (1-0.05)\mathbf{v}_i + \Delta t\, \mathbf{f}_i$, $\mathbf{x}_i \leftarrow (\mathbf{x}_i + \Delta t\, \mathbf{v}_i)/|\cdot|$ with $\Delta t = 0.3$ (velocities kept tangential) is iterated to equilibrium. The 12 icosahedron vertices are fixed. $\bar{d}$ is fixed by the level, the side of the equilateral triangle with the mean cell area, $\bar{d} = \sqrt{4\pi/(5\sqrt{3})}/2^n$, so that the equilibrium does not depend on the starting mesh. The relaxation is applied after every bisection (multilevel), and it stops when the largest tangential force is below $10^{-10}\beta\bar{d}$. For $\beta = 1$ the springs push against each other weakly; Tomita et al. recommend $\beta = 1.2$, and $\beta \geq 1.3$ folds the mesh over (the code checks the orientation of all triangles). With $\beta = 1.2$ the asymmetry drops to 0.10/0.089 at level 6/7 and the TRiSK tangential error to 0.014–0.018.

2.5 Spherical centroidal Voronoi tessellations

An SCVT is a Voronoi diagram whose generators are the centroids of their cells (Du, Gunzburger and Ju 2003); the MPAS grids are SCVTs. Lloyd's algorithm iterates $\mathbf{x}_i \leftarrow c(V_i)/|c(V_i)|$. The centroid of the spherical polygon V_i with vertices $c_1, \ldots, c_m$ (the circumcenters of the triangles around $\mathbf{x}_i$) is exact:

$$\int_{V_i} \mathbf{x}\, dA = \frac{1}{2} \sum_{k=1}^{m} \theta_k \, \frac{c_k \times c_{k+1}}{|c_k \times c_{k+1}|}, \qquad \theta_k = \angle(c_k, c_{k+1}), \tag{4}$$

which follows from Stokes' theorem applied to the constant vector fields, with $\int_V \mathbf{x}\, dA = \frac{1}{2} \oint \mathbf{x} \times d\mathbf{x}$ on the unit sphere and the great-circle arc contributing θ_k times its pole.

Plain Lloyd converges slowly (4270 iterations at level 6). The fixed-point iteration $\mathbf{x} \leftarrow G(\mathbf{x})$ is accelerated by Anderson mixing (type II): with the residuals $f_k = G(\mathbf{x}_k) - \mathbf{x}_k$ and the last $m = 16$ differences ΔX, ΔF of iterates and residuals,

$$\gamma = (\Delta F^T \Delta F + \varepsilon I)^{-1} \Delta F^T f_k, \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + f_k - (\Delta X + \Delta F)\gamma,$$

with a small regularization $\varepsilon = 10^{-12} \max \mathrm{diag}$. Level 6 converges in 167 iterations, level 7 in 8 s. The connectivity is kept, so the code checks that the triangulation is still Delaunay at the end.

2.6 Cleanup of locally refined Delaunay meshes

The Delaunay dual of the variable-resolution MPAS grid x4.163842 (92–25 km, refined around 0°N 0°E, 327680 triangles) is far less regular than an ICON grid: angles up to 94.7° (60 triangles above 80°, 3 obtuse), Voronoi (dual) edges down to 1.2 km where four MPAS cells nearly meet

in one point, and $d_{e,l}/d_e$ down to 0.003. `relax_triangles` runs the spring dynamics with a natural length per edge instead of a global one: the local size at each vertex (mean adjacent edge chord) is smoothed four times over the neighbours, and edge (i, j) gets $\beta(s_i + s_j)/2$ with $\beta = 1$. It is a cleanup, not an optimization: it stops at a force tolerance of 10^{-3} of the mean size (4 s), which removes the nearly right triangles long before the smooth modes converge. Result: maximum angle 84°, no obtuse triangle, shortest dual edge 4.7 km (four times the time step), edges 15–127 km (18–121 before), so the refinement is kept.

2.7 Kite grids

Every polygonal cell is split into kites (primal vertex, edge point, center, edge point); the kites around a primal vertex form a dual cell. The edge point is the crossing with the dual edge when the mesh has an orthogonal dual. See the kite note.

3 Compatible finite elements

3.1 Spaces

Elements of any order on triangles and quadrilaterals: `Lagrange(k)` (continuous, Gauss–Lobatto nodes on quads), `Discontinuous(k)`, `RaviartThomas(k)` (RT on triangles, RTCF on quads: normal components at Gauss points on the edges plus interior moments), `BrezziDouglasMarini(k)` on quads ($[P_k]^2 \oplus \mathrm{span}\{\mathrm{curl}(x^{k+1}y), \mathrm{curl}(xy^{k+1})\}$, the same normal traces as RT_k with fewer dofs) and Nédélec elements. The reference map is the (bi)linear interpolant of the cell vertices projected radially onto the sphere. H(div) fields use the contravariant Piola map $\mathbf{v} = J\hat{\mathbf{v}}/\det J$, so that $(\nabla\cdot\mathbf{v})\eta\,dA = (\hat{\nabla}\cdot\hat{\mathbf{v}})\hat{\eta}\,d\xi$ and $\mathbf{v}\cdot\mathbf{n}\,ds = \hat{\mathbf{v}}\cdot\hat{\nu}\,dt$ do not depend on the metric. The surface normal is $\hat{\mathbf{r}} = J_1 \times J_2/|J_1 \times J_2|$.

The degrees of freedom are attached to mesh points by a `Section`; closures give global indices with orientation signs, so normal continuity of H(div) and continuity of H^1 spaces hold exactly.

Mass solvers. `mass_solver(V)` assembles and caches the mass matrix of a space: diagonal when lumped at the element nodes (Gauss–Lobatto or Gauss–Radau tensor nodes), block diagonal for broken spaces (stored inverses of the blocks), otherwise a sparse Cholesky factorization. In parallel the coupled matrices are solved by gathering (`GatheredMassSolver`, replicated Cholesky) or with cuDSS on GPUs.

3.2 Linear shallow water

With $\mathbf{u} \in \mathrm{RT}_k$, $h \in \mathrm{DG}_k$,

$$M_u\dot{\mathbf{u}} = -C\mathbf{u} + gD^T h, \qquad M_h\dot{h} = -HD\mathbf{u}, \qquad D_{ij} = \int \psi_i \nabla\cdot\mathbf{v}_j, \quad C_{ij} = \int f\,\mathbf{v}_i\cdot(\hat{\mathbf{r}}\times\mathbf{v}_j). \tag{5}$$

C is antisymmetric, so the energy $\frac{1}{2}H\mathbf{u}^T M_u\mathbf{u} + \frac{1}{2}gh^T M_h h$ is conserved exactly; mass is conserved since $\nabla\cdot\mathrm{RT}_k \subset \mathrm{DG}_k$. Since $\nabla\cdot$ maps RT_k onto DG_k there are no spurious pressure modes and the steady geostrophic modes are represented (Cotter and Shipton 2012).

3.3 Nonlinear shallow water

The scheme of McRae and Cotter (2014), as in Gusto: $\mathbf{u}, \mathbf{F} \in \mathrm{RT}_k$, $h, K \in \mathrm{DG}_k$, $\omega \in \mathrm{CG}_{k+1}$,

$$\int \gamma\,\omega = -\int \mathbf{u}\cdot(\hat{\mathbf{r}}\times\nabla\gamma) + \int \gamma f, \qquad \int \mathbf{w}\cdot\mathbf{F} = \int \mathbf{w}\cdot h\mathbf{u}, \quad \int \phi\,K = \int \phi\,\tfrac{1}{2}|\mathbf{u}|^2, \tag{6}$$

$$\int \mathbf{w}\cdot\dot{\mathbf{u}} = -\int \mathbf{w}\cdot\Big(\frac{\omega}{h}\hat{\mathbf{r}}\times\mathbf{F}\Big) + \int(\nabla\cdot\mathbf{w})(gh+K), \tag{7}$$

$$\int \phi\,\dot{h} = -\int \phi\,\nabla\cdot\mathbf{F}. \tag{8}$$

Proposition 1. *The semi-discretization conserves mass and* $E_h = \int \frac{1}{2}h|\mathbf{u}|^2 + \frac{1}{2}gh^2$.

Proof. With the same quadrature in E_h and the right-hand sides, $\partial E_h/\partial\mathbf{u} = M_u\mathbf{F}$ (definition of $\mathbf{F}$) and $\partial E_h/\partial h = M_h(K+gh)$ (definition of K). Testing the momentum equation with $\mathbf{w} = \mathbf{F}$ removes the PV flux term ($(\hat{\mathbf{r}}\times\mathbf{F})\cdot\mathbf{F} = 0$ pointwise for any q), and the remaining terms cancel with the continuity equation tested with $\phi = K + gh$. □

The vorticity is diagnosed with the constant mass matrix of CG_{k+1}, so all mass matrices are factorized once. The nonlinear terms are kernels over the cells on bases tabulated once (`CellTables`).

3.4 Staggered kite elements

Summary of the kite note: the height lives in `KiteRadau(p)` (Q_p on the kites with Gauss–Radau nodes that include the center side: continuous inside a primal cell, broken across primal edges, diagonal lumped mass); the velocity in `KiteBrokenRT(k)` (normal continuity across the half edges inside a dual cell, broken across the dual-cell boundaries, block-diagonal mass per dual cell), `KiteBrokenBDM(k)` or `KiteDualRadau(p)` (Wess, Kapidani, Codecasa and Schöberl 2024); vorticity in CG_{k+1} with lumped mass. The divergence D and the gradient G are defined with face terms on the respective broken edges and satisfy $G = -D^T$ exactly. The conservative nonlinear variant projects $\mathbf{F} = \Pi_{V_u}(h\mathbf{u})$ and conserves mass and energy as in the proposition above; all mass matrices stay (block) diagonal, so the scheme is explicit and local.

4 Finite-volume upwind advection

`compute_advection_rhs!` is the first-order upwind scheme for a cell-wise constant tracer C with given normal velocities U_e:

$$A_i\dot{C}_i = -\sum_{e\in i} n_{e,i}\,l_e\,U_e\,C_{\mathrm{up}(e)},$$

with C_{up} the value of the cell upstream of edge e. It conserves $\sum A_iC_i$ and is monotone under the CFL condition.

5 C-grid finite volumes on Voronoi meshes: TRiSK

5.1 Geometry

`TRiSKGrid` describes a Voronoi–Delaunay pair: primal cells i (Voronoi polygons) with mass points at the generators g_i, dual cells v (the triangles of three generators around a mesh vertex x_v), and edges e with the normal $\mathbf{n}_e$ from the left cell to the right cell. Per edge: the primal length l_e (between its vertices), the dual length d_e (between the generators), the edge point m_e where both cross, and the distances $d_{e,i}$ of the generators to m_e. Kites $K_{i,v}$ (cell $i \cap$ dual cell v)

are pairs of spherical triangles (g_i, m_e, x_v); the cell areas A_i and dual areas A_v are their sums, so the kite fractions $R_{i,v} = K_{i,v}/A_i$ add up to one in every cell, which is what the TRiSK weights need.

On Delaunay meshes with obtuse triangles the circumcenter lies beyond an edge. All distances and kites are then *signed*: $d_{e,i} < 0$ for a generator beyond the edge, and spherical triangles get the signed area

$$A(a,b,c) = 2\,\mathrm{atan2}\big(a\cdot(b\times c),\ 1 + a\cdot b + b\cdot c + c\cdot a\big), \tag{9}$$

which is negative for clockwise triangles (Van Oosterom–Strackee). With signed kites the areas are exact for any Delaunay mesh.

5.2 Operators

With $n_{e,i} = \pm 1$ (outward) and $t_{e,v} = \pm 1$ (counter-clockwise around v):

$$(D\mathbf{F})_i = \frac{1}{A_i}\sum_{e\in i} n_{e,i}\, l_e F_e, \qquad (G\phi)_e = \frac{\phi_r - \phi_l}{d_e}, \qquad \zeta_v = \frac{1}{A_v}\sum_{e\in v} t_{e,v}\, d_e u_e. \tag{10}$$

With the inner products $\langle a,b\rangle_c = \sum_i A_i a_i b_i$ on cells and $\langle u,w\rangle_e = \sum_e l_e d_e u_e w_e$ on edges, $\langle DF, \phi\rangle_c = -\langle F, G\phi\rangle_e$ exactly (summation by parts), and the discrete identities $\zeta(G\phi) = 0$, $D(\text{skew gradient}) = 0$ hold as in the continuum.

TRiSK weights. The tangential velocity at edge e is reconstructed from the edges of its two cells (Thuburn et al. 2009),

$$v_e = \frac{1}{d_e}\sum_{e'} w_{e,e'}\, l_{e'} u_{e'}, \qquad w_{e,e'} = n_{e,i}\, n_{e',i}\Big(\tfrac{1}{2} - \sum_{v} R_{i,v}\Big), \tag{11}$$

with the sum over the corners of cell i passed when walking counter-clockwise from e to e'. The weights are antisymmetric, $w_{e,e'} = -w_{e',e}$, and the dual divergence of the perpendicular fluxes equals the kite average of the primal divergence, which makes the linear geostrophic modes steady.

5.3 The nonlinear TRiSK scheme

Following Ringler et al. (2010), with $F_e = h_e u_e$:

$$h_v = \frac{1}{A_v}\sum_i K_{i,v} h_i, \qquad q_v = \frac{\zeta_v + f_v}{h_v}, \qquad q_e = \tfrac{1}{2}(q_{v_1} + q_{v_2}), \tag{12}$$

$$\partial_t u_e = \frac{1}{d_e}\sum_{e'} w_{e,e'}\, l_{e'} F_{e'}\, \frac{q_e + q_{e'}}{2} - (GB)_e, \qquad B_i = K_i + g h_i, \qquad \partial_t h_i = -(DF)_i. \tag{13}$$

Proposition 2. *Let the kinetic energy be any quadratic form $K_i = \sum_e c_{i,e} u_e^2$ and the edge height*

$$h_e = \frac{2}{l_e d_e}\sum_i A_i\, c_{i,e}\, h_i. \tag{14}$$

Then the scheme conserves mass and the energy $E = \sum_i A_i(h_i K_i + \frac{1}{2} g h_i^2)$.

Proof. $\partial E/\partial u_e = 2u_e \sum_i A_i h_i c_{i,e} = l_e d_e\, h_e u_e = l_e d_e F_e$ and $\partial E/\partial h_i = A_i B_i$. Hence

$$\dot E = \sum_e l_e d_e F_e\, \partial_t u_e + \sum_i A_i B_i\, \partial_t h_i = \sum_{e,e'} l_e F_e\, w_{e,e'} l_{e'} F_{e'}\, \tfrac{q_e + q_{e'}}{2} - \langle F, GB\rangle_e - \langle DF, B\rangle_c.$$

The double sum vanishes because $w_{e,e'}\frac{q_e+q_{e'}}{2}$ is antisymmetric in (e,e') and the rest is symmetric; the last two terms cancel by summation by parts. Mass: $\sum_i A_i (DF)_i = 0$ since every edge appears twice with opposite signs. □

The PV flux conserves energy for *any* edge PV, the condition (14) links only K and h_e. For the cell kinetic energy (ICON's `e_inn_c`)

$$K_i^{\text{cell}} = \frac{1}{A_i} \sum_{e \in i} \tfrac{1}{2}\, l_e d_{e,i}\, u_e^2 \tag{15}$$

(14) gives the distance-weighted mean $h_e = (d_{e,l} h_l + d_{e,r} h_r)/d_e$.

5.4 Gassmann's PV flux

`pv_flux = :gassmann` replaces the TRiSK reconstruction by that of Gassmann (2013) with the PV usage of Gassmann (2018), as in ICON-IAP (`i_cori_method = 1`): the tangential flux at e is reconstructed from the twelve edges of the two vertices of e (the inner edges) and of their outer vertices,

$$\sum_k W_{k,e}\, F_{e_k}\, q_{p(k)} \approx q\, \mathbf{F} \cdot (\hat{\mathbf{r}} \times \mathbf{n}_e),$$

with rhombus PV $q_e = (\zeta_e + f_e)/h_e$, $\zeta_e = (A_{v_1}\zeta_{v_1} + A_{v_2}\zeta_{v_2})/(A_{v_1} + A_{v_2})$ (likewise h_e). An outer edge carries the PV of its inner edge, an inner edge that of the third edge of the shared vertex. The weights combine the triangle of the vertex and the three rhombi around it; with the metric $\mathbf{n}_k \cdot (\hat{\mathbf{r}} \times \mathbf{n}_e)$ taken at the midpoint of both edge points, $l_e d_e W_{k,e} = -l_{e'} d_{e'} W_{j,e'}$ for the pair with the same PV, so the term does no work. Unlike TRiSK it is consistent on the hexagon mesh: tangential maximum error 0.015 → 0.0016 from hexagon 3 to 6 (TRiSK 0.12); TC2 L2 errors are 2–4 times smaller. Without dissipation its PV usage lets grid-scale vorticity grow in the Galewsky jet from day 3; `:gassmann_mean` uses the same weights with the mean $(q_e + q_{e'})/2$ and stays smooth (three ghost rings).

5.5 Consistency of the reconstruction on optimized meshes

max error of v_e (solid body)	level 5	level 6	level 7
plain bisection	0.11	0.11	0.11
spring, $\beta = 1.2$	0.014	0.016	0.018
SCVT	0.0071	0.0072	0.0073

Table 1: TRiSK tangential reconstruction: inconsistent on every grid, as found by Peixoto (2016), but an order of magnitude better on optimized grids.

On spring hexagons the TC2 L2 error is 3–4.5 times and the maximum error 3–5 times smaller than on plain ones (hexagon 4–6). On SCVT hexagons TC2 is not better than on spring ones: the maximum error is dominated by the kinetic energy.

6 A consistent, energy-conserving kinetic energy

6.1 Why the cell kinetic energy is inconsistent

For a constant tangent vector $\mathbf{u}$ with $u_e = \mathbf{u} \cdot \mathbf{n}_e$, (15) gives $K_i = \frac{1}{2}\mathbf{u}^T T_i \mathbf{u}$ with the tensor

$$T_i = \frac{1}{A_i} \sum_{e \in i} l_e d_{e,i}\, \mathbf{n}_e \mathbf{n}_e^T,$$

and consistency requires $T_i = I$ on the tangent plane.

Lemma 1. *Let P be a planar polygon with area A, sides s_e of length $|s_e|$, outward normals $\mathbf{n}_e$ and midpoints m_e, and let c be any point. Then*

$$\sum_e |s_e|\, \mathbf{n}_e (m_e - c)^T = A\, I. \tag{16}$$

If p_e is the foot of the perpendicular from c to the line of s_e, $h_e = \mathbf{n}_e \cdot (p_e - c)$ its signed distance and $m_e - p_e = \tau_e \mathbf{t}_e$, then

$$\sum_e |s_e|\, h_e\, \mathbf{n}_e \mathbf{n}_e^T = A\, I - \sum_e |s_e|\, \tau_e\, \mathbf{n}_e \mathbf{t}_e^T. \tag{17}$$

Proof. By the divergence theorem $\int_P \nabla \mathbf{x}\, dA = \oint \mathbf{n}\, \mathbf{x}^T\, ds$; the left side is AI, the right side $\sum_e |s_e| \mathbf{n}_e m_e^T$ because $\mathbf{x}$ is linear on each side. Subtract $\sum_e |s_e| \mathbf{n}_e c^T = 0$. The second identity follows from $m_e - c = h_e \mathbf{n}_e + \tau_e \mathbf{t}_e$. □

For a Voronoi cell, $d_{e,i}$ is the distance from g_i to the edge point m_e, which *is* the foot of the perpendicular (the dual edge is orthogonal to the primal edge), but not the midpoint of the primal edge. So $T_i = I - \frac{1}{A_i} \sum_e l_e \tau_e \mathbf{n}_e \mathbf{t}_e^T$, where τ_e is the offset of the edge point from the primal edge midpoint. This offset is a fixed fraction of the cell size on irregular meshes and does not shrink with refinement: the error of K^{cell} is $O(1)$ and that of ∇K is $O(1/h)$. Next to the pentagons it is 2 % on SCVTs and 1.1–1.5 % on spring meshes at every resolution, and the momentum residual of the balanced TC2 state doubles with every level and sits next to a pentagon.

6.2 The vertex kinetic energy

For a dual triangle with its circumcenter x_v, the foot of the perpendicular from x_v to each side *is* the midpoint of that side: $\tau_e = 0$ in Lemma 1. The sides are the dual edges ($|s_e| = d_e$) and the distances are $l_{e,v}$, the length of the part of the primal edge from x_v to m_e. With the areas

$$A_{e,v} = \tfrac{1}{2} l_{e,v} d_e = \text{area of the triangle } (g_l, g_r, x_v), \tag{18}$$

Lemma 1 gives $\sum_{e \in v} A_{e,v} \mathbf{n}_e \mathbf{n}_e^T = \frac{1}{2} A_v I$ (exactly in the plane, up to $O(h^2)$ curvature terms on the sphere), so

$$K_v = \frac{1}{A_v} \sum_{e \in v} A_{e,v}\, u_e^2 \tag{19}$$

is consistent on *any* Delaunay triangle. The cell value is the kite average, $K_i^{\text{vertex}} = \sum_v R_{i,v} K_v$, as in Ringler et al. (2010) and Skamarock et al. (2012). Two details matter:

- MPAS uses $l_e d_e / 4$ instead of $A_{e,v}$, which assumes that x_v lies at the midpoint of the primal edge, i.e. it reintroduces an offset like τ_e. On spring hexagon 5 that version has a 3.1 % error, worse than the cell kinetic energy (1.4 %).
- On an obtuse dual triangle the circumcenter lies beyond the long side and $A_{e,v}$ for that side is negative. The sign must come from the topology, not from the position of x_v: in the counter-clockwise left cell of e, crossing an edge that ends at v turns counter-clockwise around v, and then $A_{e,v} = +A(g_l, g_r, x_v)$, otherwise $-A(g_l, g_r, x_v)$. With this sign $\sum_e A_{e,v} = A_v$ to round-off also for non-convex dual polygons (`vertex_edge_areas`).

6.3 Mixture and energy conservation

`TRiSKShallowWater(...; kinetic_energy)` uses

$$K_i = \alpha K_i^{\text{cell}} + (1-\alpha)\sum_v R_{i,v} K_v, \qquad \alpha = 1 \text{ (:cell)},\ \tfrac{3}{8} \text{ (:mixed)},\ 0 \text{ (:vertex)}. \tag{20}$$

Its coefficients are $A_i c_{i,e} = \frac{\alpha}{2} l_e d_{e,i} + (1-\alpha)\sum_{v\in e} K_{i,v} A_{e,v}/A_v$, so (14) gives the energy-conserving edge height

$$h_e = \alpha\,\frac{d_{e,l}h_l + d_{e,r}h_r}{d_e} + (1-\alpha)\,\frac{2\,(A_{e,v_1}h_{v_1} + A_{e,v_2}h_{v_2})}{l_e d_e}, \qquad h_v = \frac{1}{A_v}\sum_i K_{i,v} h_i, \tag{21}$$

the vertex part being the analogue of the distance-weighted mean along the primal edge (the weights add up to one up to curvature). The vertex heights h_v are the ones the PV needs anyway. By Proposition 2 mass and energy are conserved for every α. The stencil of K at a ghost cell next to an owned edge reaches the edges of the cells around its vertices, which are inside the two vertex rings of ghost cells the PV flux already needs; 1 and 3 MPI ranks give bitwise identical results.

6.4 Results

mesh	K	hexagon 4	hexagon 5	hexagon 6
error of K for a solid-body rotation, relative				
spring	cell	1.5 %	1.4 %	1.25 %
spring	vertex	0.60 %	0.27 %	0.13 %
SCVT	cell	2.0 %	2.0 %	2.0 %
SCVT	vertex	0.50 %	0.21 %	0.09 %
plain	cell	5.9 %	5.8 %	5.8 %
plain	vertex	0.71 %	0.33 %	0.16 %
TC2, maximum height error at $t = 1$				
spring	cell	4.9e-4	3.2e-4	2.6e-4
spring	mixed	2.6e-4	1.4e-4	1.0e-4
spring	vertex	1.8e-4	7.2e-5	3.8e-5
SCVT	cell	6.3e-4	4.2e-4	3.5e-4
SCVT	vertex	2.7e-4	9.9e-5	4.3e-5
TC2, L2 height error at $t = 1$				
spring	cell	1.4e-4	3.9e-5	1.2e-5
spring	vertex	9.0e-5	2.2e-5	5.7e-6

Table 2: Cell versus vertex kinetic energy (TRiSK PV flux). The vertex K converges at first order and lets the TC2 maximum error converge again (about first order, the remaining error is the TRiSK reconstruction); with the cell K it stalls. Energy is conserved to round-off with all variants (relative change below 10^{-14} over one time unit on spring hexagon 6 with $\alpha = 0$).

The Galewsky jet runs stably for 6 days with all three variants on plain and spring hexagon 6, and with the vertex K on SCVT hexagon 6 and spring hexagon 7: energy conserved to $5 \cdot 10^{-10}$ or better, the vorticity extremes and the roll-up unchanged, no grid-scale noise (the known risk of a pure vertex kinetic energy, the reason for Skamarock's mixture). The vertex part costs 25–35 % more time per step. The default stays `:cell`; the results argue for `:vertex`.

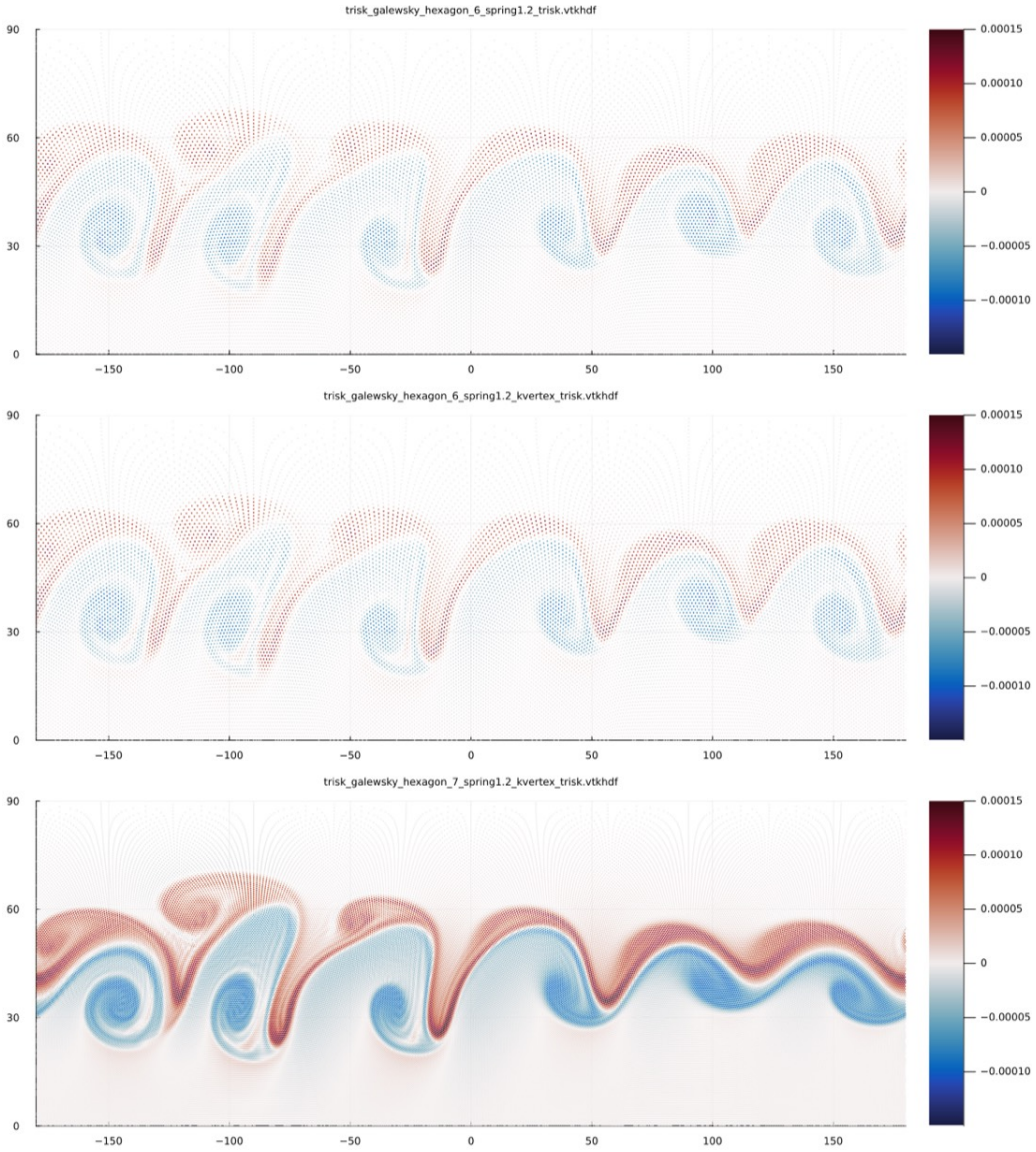


Figure 1: Galewsky jet, relative vorticity at day 6 (1/s): spring hexagon 6 with the cell (top) and the vertex kinetic energy (middle), spring hexagon 7 with the vertex kinetic energy (bottom).

7 The triangular C grid of ICON

7.1 Operators

`ICONShallowWater` ports the horizontal operators of ICON's nonhydrostatic core (`velocity_tendencies`, ICON 2024.07) to the shallow-water equations: normal velocities at the edge midpoints, heights at the triangle circumcenters, vorticity at the vertices (the hexagonal dual cells). It reuses `TRiSKGrid` with the roles of the meshes exchanged.

$$v_e = \sum_{k=1}^{4} c_{e,k} u_k, \qquad K_e = \tfrac{1}{2}(u_e^2 + v_e^2), \qquad K_i = \sum_e b_{i,e} K_e, \qquad \zeta_e = \tfrac{1}{2}(\zeta_{v_1} + \zeta_{v_2}), \tag{22}$$

$$\partial_t u_e = (f_e + \zeta_e) v_e - \big[K_e(a_1 - a_2) + a_2 K_r - a_1 K_l\big] - g\frac{h_r - h_l}{d_e}, \tag{23}$$

$$\partial_t h_i = -\frac{1}{A_i} \sum_e n_{e,i} l_e h_e u_e, \qquad h_e = \frac{d_{e,r} h_l + d_{e,l} h_r}{d_e}, \tag{24}$$

with the vector-RBF tangential velocity v_e, the bilinear edge-to-cell weights $b_{i,e}$ (`e_bln_c_s`), the linear interpolation h_e to the edge midpoint (`c_lin_e`) and $a_1 = d_{e,r}/(d_{e,l}d_e)$, $a_2 = d_{e,l}/(d_{e,r}d_e)$ (`coeff_gradekin`). The RBF coefficients of edge e use the other four edges of its two triangles, the kernel $\varphi(r) = (1 + (r/0.5)^2)^{-1/2}$ times $\mathbf{n}_i \cdot \mathbf{n}_j$, normalized so that constant vectors along $\hat{\mathbf{r}} \times \mathbf{n}_e$ are reproduced. The bilinear weights satisfy $\sum_k b_k = 1$ and $\sum_k b_k x_k = 0$ in the tangent plane, so linear fields are not aliased into a checkerboard. Mass is conserved; energy is not (the RBF reconstruction is not part of an antisymmetric operator).

7.2 The kinetic-energy gradient on irregular triangles

The gradient of ICON is the derivative of the parabola through $(-d_{e,l}, K_l)$, $(0, K_e)$, $(d_{e,r}, K_r)$ at the edge: second order, but it contains the edge's own K_e (with the RBF v_e^2) with the weight

$$a_1 - a_2 = \frac{d_{e,r}^2 - d_{e,l}^2}{d_{e,l} d_{e,r} d_e},$$

which vanishes only for symmetric circumcenters and grows like $1/d_{e,l}$ when a circumcenter approaches the edge. On the symmetric ICON grids it is harmless; on plain bisection meshes (asymmetry 0.19 at every level) it couples K_e, which contains the tangential velocity, into the momentum equation in a Hollingsworth-like way: the Galewsky jet blows up on triangle 5/6/7 before day 3/2/1, earlier on finer meshes, independent of time step and damping, and the TC2 L2 error stalls ($1.9 \cdot 10^{-5}$, $7.0 \cdot 10^{-6}$, $3.9 \cdot 10^{-6}$ for triangle 4/5/6). With the centered gradient $(K_r - K_l)/d_e$ (`kinetic_gradient = :centered`, $a_1 = a_2 = 1/d_e$) triangle 5 stays stable. On spring meshes ($\beta = 1.2$) the original scheme works: TC2 converges at second order ($2.3 \cdot 10^{-5}$, $5.6 \cdot 10^{-6}$, $1.4 \cdot 10^{-6}$) and the Galewsky jet is stable on triangle 5, 6 and 7 (energy within $3 \cdot 10^{-7}$ on triangle 7, roll-up as on R02B05).

ICON averages K_e to the cells with the bilinear weights. A cell kinetic energy of the form (15) on the triangles, $\sum_e \frac{1}{2} l_e d_{e,i} u_e^2 / A_i$ with l_e the triangle side and $d_{e,i}$ the distance of the circumcenter, would be consistent by Lemma 1, since the circumcenter projects onto the side midpoints; together with (14) it would make the mass flux part energy consistent, but the RBF Coriolis term still is not.

7.3 Divergence damping

The triangular C grid has divergence and checkerboard modes that are not controlled without dissipation; ICON adds fourth-order divergence damping $-\nu_4 \nabla(\nabla \cdot (\nabla(\nabla \cdot \mathbf{u})))$ with $\nu_4 = 0.0025\, \bar{A}^2/\Delta t$ (`divdamp_fac`, mean cell area $\bar{A}$). On locally refined meshes the time step is set by the smallest cells, and a uniform ν_4 from the mean area exceeds the explicit stability limit there, which scales with $A_{\min}^2/\Delta t$. `divergence_damping` therefore accepts one value per edge; the examples use $\nu_{4,e} = 0.0025\, A_e^2/\Delta t$ with A_e the mean area of the two cells of e.

7.4 Locally refined MPAS meshes

On the raw Delaunay dual of x4.163842 (Section 2.6) TC2 with `coeff_gradekin` blows up before $t = 0.25$, both on the raw and on the cleaned mesh. With the centered gradient it is stable: on the raw dual the L2 error is $1.2 \cdot 10^{-6}$, but the maximum error grows to $5.9 \cdot 10^{-5}$ at $t = 1$; on the cleaned mesh the L2 error is $1.2 \cdot 10^{-6}$, the maximum $1.3 \cdot 10^{-5}$, energy $-1.5 \cdot 10^{-8}$. The Galewsky jet on the cleaned mesh (6 days, $\Delta t = 5.8\,\mathrm{s}$, 89201 RK4 steps, 36 min on a laptop CPU) is NaN before day 0.5 with uniform damping and stable with the local damping: energy within $1.5 \cdot 10^{-6}$, potential enstrophy $-1.6 \cdot 10^{-4}$, $|\zeta| \le 1.44 \cdot 10^{-4}\,\mathrm{s}^{-1}$, the expected roll-up without imprint of the refinement, close to R02B06.

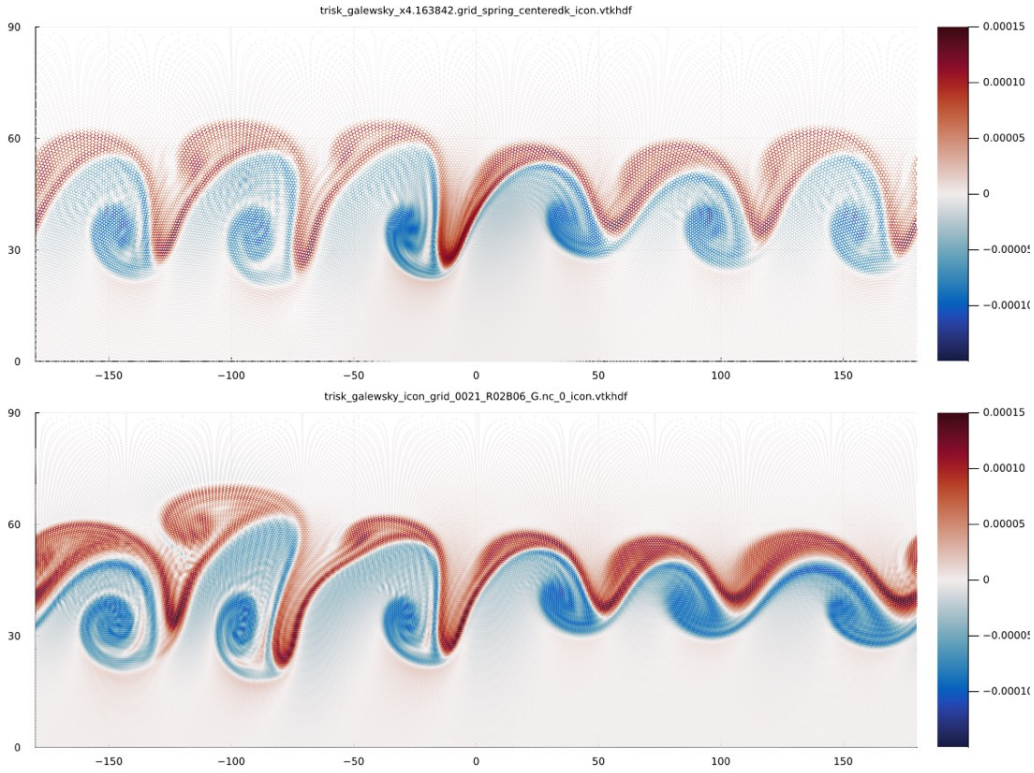


Figure 2: Galewsky jet, relative vorticity at day 6 with the ICON model: on the cleaned Delaunay dual of the locally refined MPAS grid x4.163842 (centered kinetic energy gradient, local divergence damping; top) and on the ICON grid R02B06 (bottom).

8 Time integration

All models provide `tendency!(dy, y, model)` for the state $[u; h]$ and run with the classical four-stage Runge–Kutta method (`rk4_step!`) or the three-stage SSP Runge–Kutta method of Shu and Osher (`ssprk3_step!`). RK4 contains a part of the imaginary axis ($|\lambda \Delta t| \leq 2\sqrt{2}$), which suits the oscillatory, non-dissipative semi-discretizations; the energy errors quoted above are the RK4 errors. The TC2 examples take $\Delta t = \min(0.2\sqrt{4\pi/N_c},\ 0.4 \min_e d_e)$, the second bound for locally refined meshes.

9 Parallelization and GPUs

Partition. `equal_area_partition` sorts the cells from north to south and hands them out zone by zone over the zonal equal-area regions (north cap, collars, south cap), each collar from west to east (as Atlas' `EqualRegionsPartitioner`).

Finite elements. Each rank holds its owned cells and one vertex ring of ghost cells; the owned rows of the global operators are applied after a halo exchange, the nonlinear terms are integrated over the local cells. Block diagonal mass matrices are solved locally, coupled ones by `GatheredMassSolver` or cuDSS (single GPU or multi-GPU multi-node).

C-grid models. `TRiSKDistribution` keeps two vertex rings of ghost cells (three for `:gassmann_mean`) with all their edges and vertices. Then the whole stencil of the owned tendencies is local: the state $[u; h]$ is exchanged once per tendency (one merged message per neighbour), the intermediate quantities (K, F, PV) are computed on all local entities. All operators are

KernelAbstractions kernels that gather over padded neighbour tables (no atomics), so results are bitwise identical on 1, 3 and 4 ranks, and on 1 and 4 GPUs. Per RK4 step on Levante A100: TRiSK hexagon 9 (2.6M cells) 16.7 ms on one and 5.8 ms on four GPUs; ICON R02B08 (5.2M triangles) 7.8 ms on one GPU.

10 Open questions

- Make the vertex kinetic energy the default for TRiSK, and try it together with Gassmann's PV flux (both consistent: the remaining TC2 error should then be second order also in the maximum norm).
- Energy-conserving dissipation for the C-grid schemes: APVM or an upwind PV, hyperdiffusion scaled with the local cell size on refined meshes.
- TRiSK on the locally refined MPAS Voronoi mesh itself with the vertex kinetic energy, and the refined region placed under the Galewsky perturbation (rotated grid).
- Distributed mesh construction (the global grid and the tables are still built on every rank).

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